

Program Output

Max Heap vs Linear Search

MAX HEAP INSERTIONS

After inserting 78: 78 | Comparisons: 0

After inserting 92: 92 78 | Comparisons: 1

After inserting 65: 92 78 65 | Comparisons: 1

After inserting 88: 92 88 65 78 | Comparisons: 2

After inserting 95: 95 92 65 78 88 | Comparisons: 2

After inserting 72: 95 92 72 78 88 65 | Comparisons: 1

After inserting 84: 95 92 84 78 88 65 72 | Comparisons: 1

After inserting 90: 95 92 90 78 88 65 72 84 | Comparisons: 2

Final Max Heap: 95 92 90 78 88 65 72 84

MAXIMUM USING MAX HEAP

Highest score: 95

Comparisons: 0 (root access)

MAXIMUM USING LINEAR SEARCH

Highest score: 95

Comparisons: 7

TOTAL INSERTION COMPARISONS: 10